Content Generation for Information Retrieval Systems: A Survey from the Supply-Side Optimization Perspective

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Abstract

With the rapid growth of commercial content platforms (e.g., TikTok, YouTube), modern information retrieval (IR) systems have evolved into two-sided ecosystems with both supply sides (e.g., content creators) and demand sides (e.g., users). From an economic perspective, traditional IR research primarily focuses on optimizing supply-demand matching through IR models while treating content supply as exogenous and fixed. However, such a design may pose risks of market failure in two-sided platforms, leading to degraded user experience and severe Matthew effect. Recently, content generation techniques powered by generative models have emerged as a potential solution to such risks by enabling high-quality and scalable supply-side optimization. However, it also introduces new challenges to IR systems, including difficulties in evaluation and potential risks to the content ecosystem.

In this paper, we provide a comprehensive survey of recent advances in content generation for information retrieval (IR) systems from the perspective of economic supply-side optimization. Our review is structured along three key dimensions: methods, evaluation, and challenges. Specifically, grounded in a two-sided market framework, we categorize existing content generation approaches into two broad classes according to their demand-side optimization objectives: generic content generation and personalized content generation. Next, we systematically review evaluation strategies for content generation in IR systems, progressing from high-level objectives and methodological paradigms to fine-grained dimensions and associated metrics. Finally, we identify key challenges and highlight promising future directions, with the goal of advancing the understanding and development of content generation for supply-side optimization in IR systems. We also consistently maintain a GitHub repository for the relevant papers and resources in this rising direction at https://github.com/shawnye2000/Content_Generation_in_IR_Survey.

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CCS Concepts

• **Information systems** → **Multimedia content creation**.

Keywords

Content Generation, Supply-Side Optimization, Generative Models

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1 Introduction

Two-Sided Information Retrieval Systems. Over the past two decades, the rapid expansion of commercial content platforms (e.g., TikTok, YouTube) has transformed modern information retrieval (IR) systems into complex, two-sided ecosystems consisting of supply and demand sides [1, 13]. Within such ecosystems, the supply side (e.g., content creators, advertisers) provides content (e.g., advertisements, notes, short videos), while the demand side (i.e., users) interacts with them through algorithmic ranking and exposure mechanisms driven by IR models. Traditional IR research [12, 128] typically focuses on the optimization of supply-demand matching driven by IR models. It treats content as exogenous and static, relying on retrieving items from a fixed corpus and allocating the relevant content to users to improve user experience [7, 22, 100].

Supply-Side Optimization for IR Systems. However, in such two-sided IR systems, user experience is influenced not only by the relevance of retrieved contents, but also by their intrinsic quality (e.g., attractiveness, fidelity, alignment with user preference) [107, 121]. From an economic perspective, optimizing only

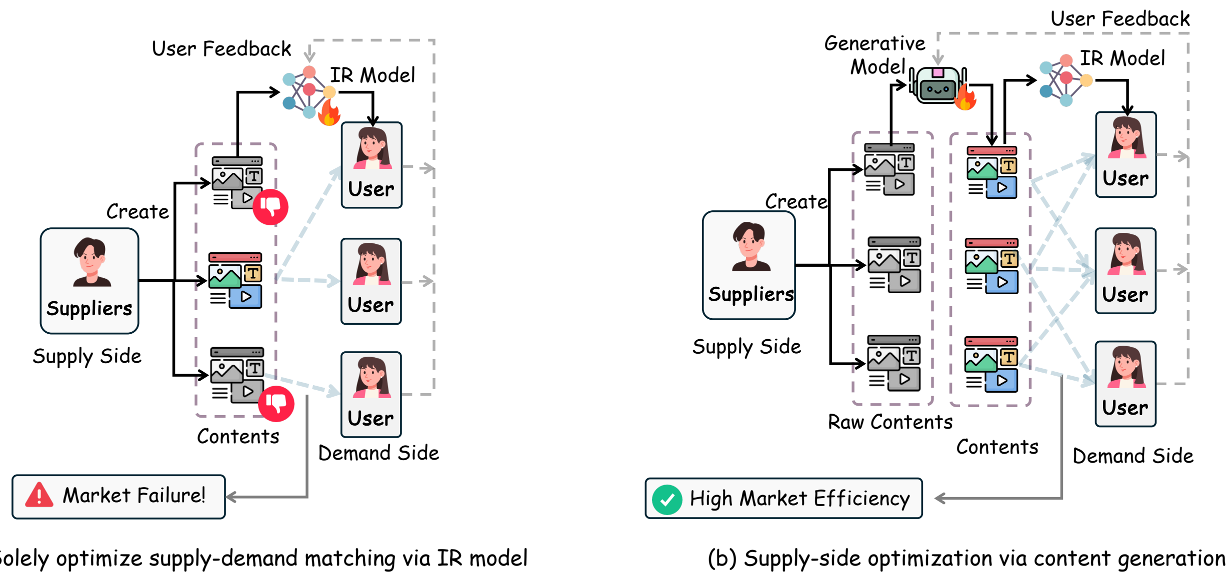


Figure 1: (a) Traditional IR systems typically treat content supply as static and exogenous, focusing on improving the matching between content supply and user demand, which may lead to suboptimal allocation and inefficiency due to low-quality content supply. (b) Content generation powered by generative models offers new opportunities to optimize content supply, thereby improving market efficiency.

supply-demand matching (i.e., optimizing IR models over a fixed content pool) may lead to inefficient allocation and even **market failure** [97, 100]. This is reflected in the declining capabilities across different sides of the IR system: severe filter bubbles and degraded user experience and engagement, strong **Matthew effects** [91] and supply-side unfairness [28], as well as a proliferation of low-quality content (e.g., clickbait) within the ecosystem.

Therefore, recent literature in IR systems [11, 82] has increasingly recognized that **supply-side optimization** is indispensable in two-sided platforms. Unlike supply-demand matching [22], which focuses on allocating existing content, supply-side optimization targets the generation and improvement of high-quality content for future distribution. Such optimization offers multiple benefits: (1) higher-quality content improves downstream ranking and leads to better user satisfaction and engagement; (2) it enhances supplier-side outcomes (e.g., exposure, revenue), thereby incentivizing sustained participation and higher-quality creation; and (3) it improves overall ecosystem health by mitigating the proliferation of low-quality or misleading content. To better illustrate the distinction between traditional IR and supply-side optimization, we provide a concrete example in Figure 1 and Table 1.

Content Generation for Supply-Side Optimization. With the rise of generative models [143, 156], content generation has become a transformative paradigm for supply-side optimization in IR systems, fundamentally reshaping the way content is produced beyond traditional human-crafted pipelines [15, 36, 50]. In this context, contents in IR systems (e.g., advertisements and item descriptions) can be produced and optimized in a more automatic, scalable, and personalized manner, significantly improving both the quality and efficiency of supply-side optimization. Despite these opportunities, a fundamental misalignment between the objectives of content generation and IR system introduces significant challenges in both evaluation and its impact on the content ecosystem.

Recently, a growing body of research has explored content generation with generative models in two-sided IR systems. However, the existing literature remains highly fragmented and siloed, with significant disparities in generation techniques, evaluation approaches, and challenge characterization. It lacks a unified framework to integrate these aspects with the objectives of IR systems. Consequently, a systematic and coherent understanding of content generation in two-sided IR systems is still missing. This fragmentation impedes the prospective research in supply-side optimization required to foster a robust, sustainable, and high-quality two-sided IR ecosystem.

Organization. To this end, our survey aims to provide a comprehensive and unified economic supply-side optimization perspective that systematically summarizes the methods, evaluation paradigms, and key challenges of content generation in two-sided IR systems. Our survey begins with a brief introduction to two-sided IR systems and the formulation of the optimization paradigms of IR systems (i.e., supply-demand matching and supply-side optimization), thereby laying the foundation for understanding how content generation can be integrated into two-sided IR systems. Based on the foundations, our survey primarily addresses three research questions:

- **RQ1: What are content generation methods in two-sided IR systems?** To address this question, we adopt an economic perspective to develop a taxonomy of existing content generation methods, categorizing them into two groups based on their demand-side optimization objectives: generic user demand and

Table 1: Comparison between supply-demand matching optimization and supply-side optimization in IR systems.

Dimension	Supply-Demand Matching Optimization	Supply-side Optimization
Optimization target	Information retrieval models	Content generation models
Objective	Improve user engagement	Improve user engagement, enhance supplier outcome
Approach	Allocation of existing content from the corpus	New content creation and existing content refinement
Execution stakeholder	Platform	Platform, content supplier
Typical operations	User modeling, intent understanding	Personalized content generation, multi-modal alignment
Evaluation dimensions	Matching accuracy	Fidelity, diversity, compatibility, etc

personalized user demand. Detailed discussions can be found in Section 3 and Section 4.

- **RQ2: How to evaluate content generation in two-sided IR systems?** In this research question, we comprehensively review how existing studies evaluate the utility of content generation in IR systems, considering its alignment with both generic and personalized user demand. We present a systematic, coarse-to-fine taxonomy spanning evaluation dimensions to specific metrics. Details can be found in Section 5.
- **RQ3: What challenges and future directions does content generation have in two-sided IR systems?** We systematically review the key challenges introduced by generative supply-side optimization and outline promising future directions. This analysis provides a holistic view of the landscape and highlights pathways toward the effective and trustworthy integration of generative models into the supply side of IR systems. Detailed discussions are presented in Section 6.

Difference with Existing Surveys. A fast-growing body of surveys has begun to chart LLMs, foundation models, and generative models for IR systems, but with different emphases. Broadly, existing works fall into two groups: (i) surveys that investigate how generative models enhance supply-demand matching (i.e., IR model), including approaches based on improved IR models [73, 120, 128] as well as generative IR frameworks [40, 63]; and (ii) surveys that focus on leveraging generative models (e.g., LLMs, diffusion models) to generate or edit content (e.g., text, images, or video artifacts) [137, 159], but without grounding these generative techniques into IR system applications (e.g., recommendation, advertising, search). As shown in Table 4, we provide a visual and tabular comparison to highlight the key differences between our survey and existing surveys. Our survey examines content generation techniques in IR systems from an economic supply-side optimization perspective, taking into account diverse IR scenarios and applications, and systematically characterizing their development, evaluation, and the challenges they introduce to the content ecosystem.

Summary of Contributions. Building on the above view, our main contributions are:

- We adopt a novel economic supply-side optimization perspective as a unifying lens to systematically organize, review, and taxonomize content generation research in IR systems.
- We systematically organize existing evaluation dimensions and metrics to characterize how generated content is assessed within IR systems, and provide a unified synthesis of the key challenges and future directions in integrating content generation into IR ecosystems.
- We identify the future directions, providing insights to facilitate the development of this potential and demanding research area.

2 Foundations and Definitions

In this section, we first introduce two-sided IR systems. We then formulate the optimization paradigms in two-sided IR systems to highlight their differences and lay the foundation for a taxonomy of content generation in IR systems. Finally, we outline the structure of this survey.

2.1 Two-sided Information Retrieval System

With the recent rise of online commercial platforms (e.g., TikTok, Xiaohongshu), traditional IR systems have undergone significant transformations and gradually evolved into two-sided systems [100]. A two-sided IR system is a multi-stakeholder, multi-stage content flow system centered around content and the services that support its retrieval [1, 11]. Next, we introduce the key stakeholders and the content flow pipeline within the ecosystem.

2.1.1 ***Two-Sided IR System.*** Recent research on IR systems emphasizes that a content-oriented IR system operates as a dynamic environment of interacting participants that evolves over time and typically involves three key stakeholders [1, 11]:

- **Supply-Side** refers to the entities responsible for producing and optimizing content within the system. It includes content creators, advertisers, platform-side content generation models, etc. Their primary responsibility is to produce content and provide it to the demand side for consumption. Their objective is to optimize content to maximize content exposure, user engagement, or monetization.
- **Platform** functions as the intermediary that connects the supply and demand sides and facilitates their interaction. It mainly consists of information retrieval, ranking, and recommendation models. Its primary responsibility is to match generated content from the supply side with user needs from the demand side. The platform aims to improve the efficiency and effectiveness of this matching process, thereby optimizing overall system utility.
- **Demand (User)-Side** refers to the consumers of content who seek to satisfy their information needs, preferences, or goals. Users interact with the system through behaviors such as clicks, views, and dwell time. Their primary objective is to discover relevant and satisfying content under limited attention. In modern platforms, the attention they allocate has become not only an optimization objective for the platform's information distribution but also a goal for supply-side optimization.

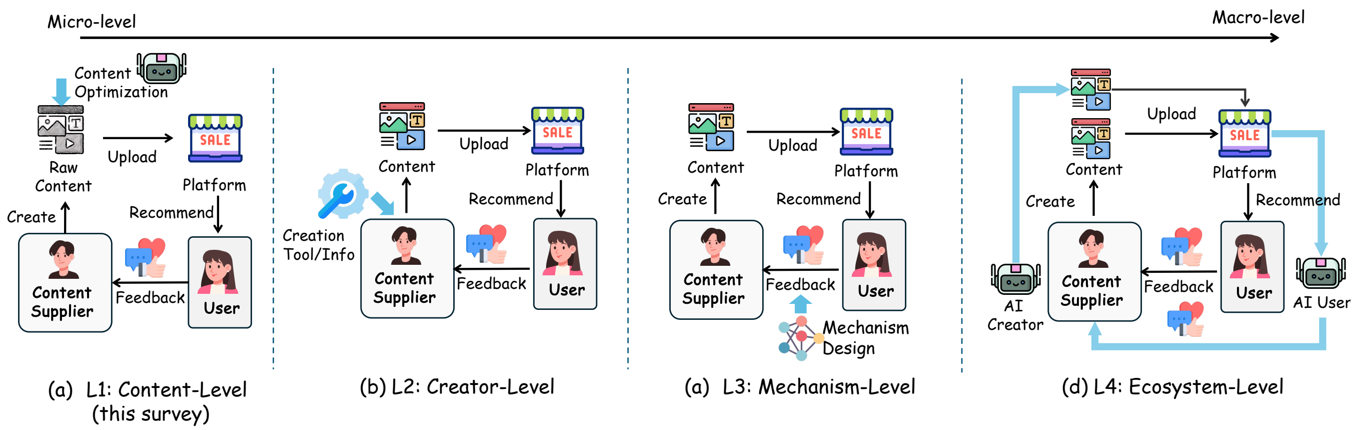


Figure 2: Four levels of intervention in supply-side optimization for IR systems, ranging from micro-level content manipulation (L1) to macro-level ecosystem design (L4). This survey primarily focuses on content-level (L1) interventions.

2.2 Optimization in Two-Sided IR Systems

In this section, we formally characterize demand-side and supply-side optimization in two-sided IR systems, with the goal of clarifying their fundamental differences and laying the groundwork for our subsequent taxonomy of generative supply-side optimization.

• **Supply-Demand Matching Optimization.** The supply-demand matching problem can be formulated as learning a parameterized matching function (i.e., information retrieval policy) π_θ, which maps each user $u \in \mathcal{U}$ to a distribution over items in a fixed corpus $\mathcal{I}$. The objective is to maximize overall user satisfaction:

$$\theta^* = \arg\max_{\theta} \sum_{u \in \mathcal{U}} \mathbb{E}_{i \sim \pi_\theta(\cdot|u)} \left[\mathcal{R}(u, i)\right]$$

where $\mathcal{R}(u,i)$ denotes the reward obtained from user u when interacting with item i. $\pi_\theta(\cdot \mid u)$ defines a distribution over items in the corpus $\mathcal{I}$, from which items are retrieved (or ranked) for user u. The interaction between a user u and an item can be described in two ways:

- **Traditional Information Retrieval Engine** [162]: The platform directly presents retrieved content to users by applying the IR model $\mathcal{R}(\cdot)$ over the content pool $\mathcal{I}$. In such way, the user reward $\mathcal{R}(u,i)$ is typically defined based on users' immediate behaviors, such as implicit (e.g., clicks, dwell time) and explicit feedback (e.g., ratings).
- **Generative Information Retrieval Engine** [63]: The retrieved items is used as input context to a generative model (e.g., LLMs), which generates a response presented to the user: In generative IR systems [63], the reward $\mathcal{R}(u,i)$ is often quantified by the quality of the generated response, for example, its correctness with respect to ground truth [40], or via a learned reward model aligned with human preferences [43].

• **Supply-Side Optimization.** Supply-side optimization focuses on improving or reshaping the content supply in IR systems beyond static retrieval over fixed corpora. As shown in Figure 2, supply-side optimization paradigms can be categorized into four levels of intervention:

- **L1: Content-Level Intervention.** Directly operates on content artifacts via generation or enhancement. This includes generic content enhancement (e.g., rewriting or enriching existing items), content personalization (user-specific content generation), and end-to-end content synthesis that aims to fully automate content creation.
- **L2: Creator-Level Intervention.** Aims to support human creators through AI systems, including creator co-pilot tools [33, 98] for content creation, as well as information design approaches [92, 157] that expose platform signals or user preference distributions to guide content creation.
- **L3: Mechanism-Level Intervention.** Focuses on system-level optimization of supply outcomes via algorithmic or economic mechanisms, including traffic allocation strategies [88, 132–134, 149], incentive mechanism design [49, 147], and welfare-oriented optimization [82] of platform objectives.
- **L4: Ecosystem-Level Intervention.** Considers the interactions among heterogeneous agents (e.g., human creators, users, generative models, and platforms) [146], emphasizing long-term ecosystem dynamics such as fairness, competition, content diversity, and sustainability.

These four levels of intervention range from micro-level content manipulation to macro-level ecosystem design, with increasing complexity and implementation difficulty. In this work, we primarily focus on L1 (content-level) interventions, while leaving other levels (L2-L4) for future work.

2.3 Content Generation for Supply-Side Optimization

Instead of treating items i as static and passively waiting to be retrieved, content-level intervention (i.e., content generation) views each item (e.g., copy, image, video) as an optimizable content entity, and focuses on improving the quality of the content supply itself. Let c denote the optimized version of item i, and $\mathcal{Z}(\cdot)$ is the utility function of the optimized content c. We taxonomize existing content generation paradigms for supply-side optimization into two representative categories:

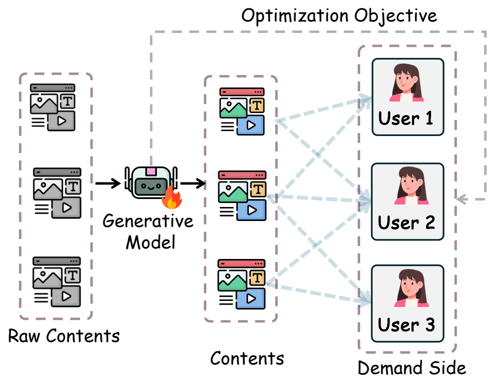


Figure 3: Generic content generation.

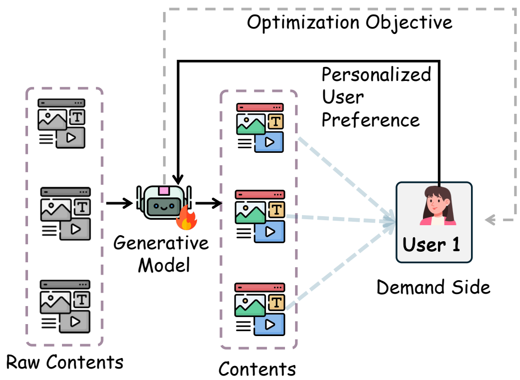


Figure 4: Personalized content generation.

- **Generic content generation**: As illustrated in Figure 3, this paradigm optimizes content conditioned on an existing item i (e.g., advertising image generation, copy rewriting), aiming to improve its quality and align with generic user demand before matching. We term this paradigm "generic optimization" because it is not tied to any specific user. The objective can be formulated as:

$$\gamma^* = \arg\max_{\gamma} \sum_{i \in \mathcal{I}} \mathbb{E}_{c \sim \pi_\gamma(\cdot|i)} \left[\mathcal{Z}_g(c, i) \right],$$

where $\pi_\gamma(\cdot|i)$ refines or edits the original content associated with item i. Note that the conditioning input i to $\pi_\gamma(\cdot|i)$ is not restricted to a fully specified item. It may also consist of intermediate or abstract representations, such as high-level ideas, prompts, or semantic descriptions, which are subsequently transformed into concrete content through generative models.
The generic utility function $\mathcal{Z}_g(c, i)$ evaluates the alignment between the generated content and generic user demand (i.e., human preference or group-level user preference). It not only includes the quality of the optimized content c (e.g., aesthetics, coherence, or downstream user engagement), but also its trustworthy characteristic (e.g., hallucination) to the original item i.
- **Personalized content generation**: As illustrated in Figure 4, this paradigm optimizes content presentation at serving time by explicitly conditioning on both the item and user request (i.e., implicit behavior or explicit query), thereby tailoring the same underlying item to heterogeneous user preferences (e.g., personalized outfit generation). Unlike generic optimization, which focuses on item-level content refinement, this paradigm typically operates in the user-facing stage of the IR pipeline. The objective can be formulated as:

$$\gamma^* = \arg\max_{\gamma} \sum_{u \in \mathcal{U}} \sum_{i \in \mathcal{I}} \mathbb{E}_{c \sim \pi_\gamma(\cdot|i,u)} \left[\mathcal{Z}_p(c, i, u) \right]$$

where $\pi_\gamma(c \mid i, u)$ denotes a user-conditioned content generation policy that produces a personalized version of item i for user u. $\mathcal{Z}_p(c, i, u)$ extends $\mathcal{Z}_g(c, i)$ by further incorporating personalization utility, including user-specific satisfaction, engagement, or task success.

2.4 Organization

In this work, we provide a systematic overview of existing research on content generation in IR systems. The organization of content generation in two-sided IR systems is shown in Figure 5.

• To answer RQ1, we categorize prior studies into two main classes: generic and personalized content generation. Generic content generation focuses on optimization in both traditional and generative IR systems, while personalized content generation targets user-specific preferences derived from implicit or explicit signals.

• To answer RQ2, we further review the evaluation methodologies for both categories of content generation.

• To answer RQ3, we discuss the key challenges in these settings. We then summarize future directions for content generation in IR systems, highlighting promising avenues for further research.

3 Generic Content Generation

Generic content generation primarily operates prior to user information requests, primarily aiming to optimize the outcome of content suppliers (e.g., exposure, revenue). After optimization, the resulting content is fed into downstream information retrieval systems, including both generative IR systems and traditional IR systems. Accordingly, we categorize existing approaches based on the type of downstream IR system, namely, content generation for traditional IR systems and content generation for generative IR systems. As illustrated in Figure 6, we provide an overview of the generic content generation workflow in IR systems.

3.1 Optimization for Traditional IR Systems

Generic supply optimization for traditional IR systems mainly consists of two aspects: advertising content generation and item description completion.

3.1.1 ***Advertising Content Generation.*** As a key direction of content generation for supply-side optimization, advertising content generation aims to produce faithful and engaging advertisements using generative models. In the following, we introduce optimization methods for advertising text, images, and videos, respectively.

Advertising Text Generation. Existing research on advertising text generation has evolved from neural sequence modeling to pre-trained language model–based approaches with enhanced optimization mechanisms.

Table 2: Comparison between generic and personalized content generation paradigms.

	Generic content generation	Personalized content generation
Supply involvement	Supply-independent	Supply-demand collaboration
Required information	Reference information	User preferences, Reference information
Execution stakeholder	Content supplier	Platform
Served Stakeholder	Content supplier	User
Preference objective	Population-level preference	Individual-level preference
Stage	Before information retrieval	After/without information retrieval

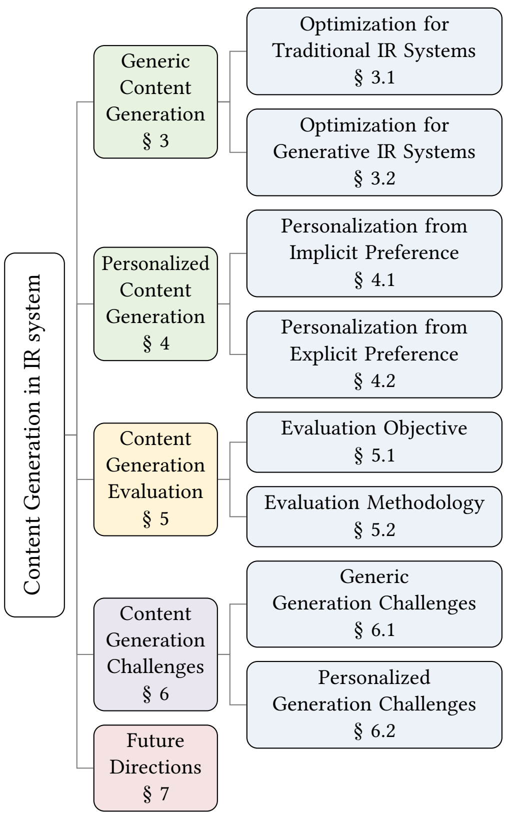


Figure 5: Organization of this survey.

Early work primarily relies on sequence-to-sequence frameworks. For instance, Hughes et al. [52] combines parallel LSTM encoder–decoder architectures with self-critical sequence training to generate high-CTR advertising text, while Kamigaito et al. [55] further introduces reinforcement learning (RL) to improve the diversity of generated content. With the emergence of pretrained language models, subsequent studies focus on improving generation quality by incorporating richer contextual signals and optimization objectives. For example, Wang et al. [123] builds upon pretrained language models [29] and leverages landing page (LP) information to generate more fluent and attractive advertising text. Furthermore, Kamigaito et al. [54] formulates ad generation as a RL problem, using LP information as a reward signal to improve both relevance and content quality.

Recent studies explore more advanced generation paradigms such as self-iterative refinement and multi-agent collaboration. For instance, GCOF [158] adopts a self-iterative framework that integrates LLMs with genetic optimization to progressively refine marketing copy. Similarly, CARTS [18] employs a multi-agent framework to generate item titles based on metadata and iteratively improve them through agent collaboration.

Beyond the aforementioned studies, several works explicitly consider attractiveness in text generation, which is often measured by click-through rate (CTR), to improve user engagement and overall revenue. Early work such as Mishra et al. [80] leverages large-scale A/B testing data to learn transformation patterns from low-CTR to high-CTR creatives. More recent approaches increasingly adopt LLMs as generators and incorporate advanced optimization techniques, including iterative refinement [113], PPO [101] with multi-reward objectives [70], online DPO [23], and contrastive fine-tuning with online A/B test feedback [127]. These methods aim to maximize click and conversion performance while preserving the novelty of advertising copy.

Advertising Image Generation. In the domain of advertising visual content generation, existing research has evolved along a trajectory of increasing automation.

Early works primarily focused on local image editing in advertising scenarios. For instance, Thomas and Kovashka [110] improves advertisement effectiveness by adjusting key visual elements. However, such approaches still rely on manually provided base materials and layouts. With the advancement of generative models, research has gradually shifted from local editing to holistic content generation. For example, Vashishtha et al. [112] combines LLMs with diffusion models to directly generate personalized banner advertisements from structured user and product features. Furthermore, Gao et al. [38] and Li et al. [62] render textual elements directly onto generated images, enabling joint visual-text generation and producing more complete advertising posters. Building upon this progress, automation has further advanced toward end-to-end poster generation. Earlier methods still require users to provide images of predefined sizes and promotional text in advance [44, 111, 144]. In contrast, AutoPoster [67] and T-Stars-Poster [16] achieve a fully automated pipeline, covering background generation, layout design, and final rendering, and are capable of producing near-commercial-quality posters.

More recent work has pushed this trend toward more structured and systematized generation. Many approaches adopt a modular design, decomposing the task into sub-processes (e.g., background generation), layout planning, and text rendering, as seen in Li et al.

[62], Prompt2Poster [119], and PosterMaker [39]. Within this framework, some studies further optimize specific modules. For instance, Product2IMG [14] enhances background generation by incorporating multimodal large language models (MLLMs) as feedback during training, while BannerAgency [116] employs multiple MLLM-based agents to collaboratively perform advertisement goal analysis, background generation, element planning, and final rendering, further improving automation and generation quality. Although automated content generation has become highly capable, images generated directly by generative models often have limited practical usability in advertising scenarios. To address this issue, CAIG [20] explicitly penalizes background–product mismatches during training, improving both advertisement CTR and visual consistency between products and generated backgrounds.

Advertising Video Generation. Beyond text and static images, recent work has extended advertising content generation to the video domain. VC-LLM [93] leverages raw visual materials and product information to automatically generate short-form advertising videos. Specifically, it performs clip selection, temporal reordering, voice-over script generation, and subtitle segmentation, enabling an end-to-end pipeline for video advertisement creation.

3.1.2 *Item Description Completion*.

Item descriptions in recommender systems are crucial for providing concise, informative summaries that attract potential users. However, in real-world production environments, these features often suffer from missing or incomplete attributes [47], and manually collecting or completing them (e.g., via web crawling or annotation) can be extremely time-consuming and costly [2]. To address this challenge, several studies leverage the world knowledge embedded in LLMs to automatically infer and complete missing item attributes.

Early works mainly focus on automatic attribute completion and structured description generation. Luo et al. [78] leverages LLMs to automatically complete and standardize product attributes, correcting missing or inconsistent information in catalog entries. Similarly, Acharya et al. [2] generates structured item descriptions from product names or metadata, which can replace or augment sparse or noisy textual fields in recommendation models, achieving performance comparable to manually curated descriptions.

3.2 Optimization for Generative IR Systems

Beyond enabling supply-side optimization for traditional retrieval- and ranking-based IR systems, generative IR systems also require dedicated supply-side optimization. In this survey, we systematically review content generation techniques tailored for generative IR systems, with a particular focus on approaches such as generative engine optimization (GEO).

Generative Engine Optimization (GEO). Traditional Search Engine Optimization (SEO) aims to improve a webpage's visibility in search engine result pages (SERPs) by aligning its signals with ranking mechanisms [83, 96, 103]. However, LLMs are shifting search engines from "returning a list of links" to "directly generating answers with citations". While this improves user experience, it significantly reduces creators' control over how and when their content is displayed, weakening exposure for websites and content providers [3]. Traditional SEO, which focuses on keyword matching and ranking, is no longer sufficient in this multi-source answer generation paradigm, motivating the need for a new optimization framework tailored to generative search.

To this end, Aggarwal et al. [3] introduce a new task and evaluation framework, Generative Engine Optimization (GEO), defined as rewriting webpage content to improve its visibility in generated answers. Through heuristic strategies, they find that generative engines favor content that is evidence-based, well-cited, information-dense, and clearly expressed. To move beyond heuristic optimization, AutoGEO [129] further identifies preference rules by comparing high-visibility and low-visibility documents, then automatically extracts and refines these rules through a four-stage pipeline. Based on these rules, two models are developed: AutoGEO$_{\text{API}}$, which directly applies rules via prompting, and AutoGEO$_{\text{Mini}}$, which uses GRPO [102] with rewards for visibility improvement, rule compliance, and semantic consistency. However, these approaches rely on static rules and cannot adapt to the rapidly evolving nature of generative search engines. To address this limitation, recent efforts turn to agentic frameworks that enable dynamic, adaptive optimization. AgenticGEO [151] continuously adapts optimization strategies through online feedback, multi-agent collaboration, and MAP-Elites–based rule evolution.

4 Personalized Content Generation

Generic content generation mainly focuses on intrinsic content quality and cannot well satisfy specific user demands in IR systems. Personalized content generation alleviates this issue by explicitly optimizing the content via demand-side signals (e.g., user preferences and queries), enabling more adaptive, controllable, and user-aligned content creation.

4.1 Personalization from Implicit Preference

Personalized content generation dynamically adapts retrieved target items based on users' implicit signals (e.g., interaction history) and behaviors, producing content that better aligns with individual preferences and intent, thereby enhancing user experience. In this part, existing works mainly adopt a *personalized generation over supplier* paradigm (Figure 7(a)).

Personalized Generic Image Generation. Several studies in recommendation scenarios encode user interaction history into personalized tokens [137], explicit keywords, or soft preference embeddings [105], which are then used to condition diffusion models to generate images that reflect both item relevance and individual user preferences. For example, I-AM-G [124] further improves cross-modal alignment by rewriting user interests, retrieving relevant historical items, and conditioning generation on multiple textual and visual sources. Recent efforts [122] also exploit conversational preference signals to guide personalized visual synthesis from dialogue-derived user inclinations. However, such approaches rely heavily on LLMs, which often introduce high inference latency and lead to text-dominant bias during generation. To address this limitation, Patron [89] demonstrates that high-fidelity, preference-aligned images can be generated using implicit feedback alone. Thus, they train a conditional diffusion prior in the CLIP embedding space based solely on users' implicit behaviors. Complementarily, RAGAR [71] improves preference alignment by retrieving

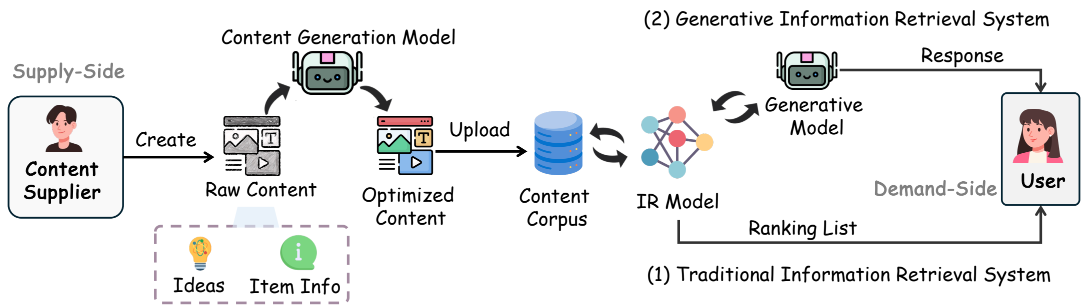


Figure 6: Generic content generation in two-sided IR systems can be applied to both traditional and generative IR systems for optimization.

semantically relevant historical items and recommender-based reflective feedback to optimize personalization beyond image-level consistency. Moreover, when combining user preferences with semantic instructions, generative models often overly rely on a single guidance signal. To mitigate this issue, DRC [140] disentangles semantic and style representations, orthogonally fusing user preferences with reference content to improve both style preservation and semantic fidelity.

Personalized Outfit Image Complement. Beyond generic image generation, an increasing number of on-demand generation studies have been applied to fashion outfit recommendation scenarios, aiming to generate compatible fashion outfits. Early work primarily rely on conditional GANs as base models to generate personalized clothing images or compatible fashion items. DVBPR [56] integrates Siamese CNN-based visual representation learning to generate user-specific fashion items, while c+GAN [59] extends this approach to complementary fashion recommendation, producing coherent item sets. MrCGAN [106] further advances compatibility modeling by representing each item with embeddings and a set of compatibility prototypes, allowing generation of semantically compatible items. Similarly, Yang et al. [145] leverage visual, semantic, and compatibility cues to produce novel outfits aligned with user preferences, improving both recommendation quality and content generation.

More recent approaches adopt advanced architectures (e.g., diffusion models and MLLMs) to better capture user intent and outfit compatibility. M6-Rec [26] uses transformers to generalize recommendation to multi-task, open-domain scenarios, transforming user behavior logs into natural language to generate new item titles, which are subsequently visualized by M6-UFC [155]. Following this line, DiFashion [136] employs diffusion models conditioned on historical context to generate compatible outfits. However, such methods often produce limited diversity in generated content. To address this, FashionDPO [150] fine-tunes pre-trained generative models via Direct Preference Optimization (DPO) [95] with multi-expert feedback, significantly enhancing diversity in generated outfit.

Personalized Text Generation. Recent studies have also explored personalized text generation for content creation. For instance, PNG [5] personalizes news headline generation by integrating multi-dimensional user profiles, knowledge-aware encoding, and a user-perturbed pointer-generator. HLLM-Creator [17] employs a hierarchical LLM framework with item, user, and creative LLMs to generate personalized ad titles conditioned on user behavior and product constraints.

Personalized Video Generation. With the advancement of generative video models, personalized video advertising is shifting from static retrieval over finite creative inventories to dynamic, on-demand synthesis. NextAds [138] proposes a closed-loop framework with four agents that respectively handle planning, generation, evaluation, and iterative refinement. By using a structured creative plan as an intermediate interface, it enhances controllability and personalization while enabling continuous optimization.

4.2 Personalization from Explicit Preference

Personalized content generation based on explicit feedback can be broadly categorized into two directions: search advertisement generation and on-demand product generation. The former typically follows a *personalized generation over supplier* paradigm (Figure 7 (a)), where generation is conditioned on existing suppliers to produce targeted advertisements, while the latter adopts a new paradigm of *personalized generation for supplier* (Figure 7 (b)), where content is directly tailored and generated for suppliers to enable content customization.

Search Advertisement Generation. Advertisement generation in search engines requires not only high quality in terms of appeal and fluency, but also being tailored to user queries [123]. This requires the generative model to generate content in real time when the user submits a query. Hughes et al. [52] introduces user query as a critical feature for the CTR oracle model in reinforcement learning, enabling reward calculation tied to user search context. Wang et al. [123] formalizes user query as an optional but dedicated input component that explicitly models user tasks and builds upon UniLM [29] and adopts a multi-stage training strategy to produce more attractive and higher-quality advertising copy. Most recently, DeepGen [41] further leverages user query for online ad assembly, where pre-generated ad fragments are dynamically selected and stitched in response to real-time user queries to highlight product aspects matching user intent.

On-Demand Product Generation. Beyond the academic studies discussed above, recent industrial practice has also begun to explore this direction. In particular, PerFusion [66] introduces a "sell-before-you-make" workflow that generates high-fidelity product images tailored to crowd-level preferences, promotes products prior to manufacturing, and determines production decisions based on observed market feedback. Compared with human-designed products, this line of work demonstrates consistent improvements in both engagement and conversion-related metrics, as well as reductions in post-purchase returns, highlighting the transformative potential of AI-generated items for e-commerce platforms. Although research in this direction remains limited, it represents a promising avenue for future exploration in personalized content generation.

5 Evaluation of Content Generation

In this section, we systematically review the evaluation methods for generative supply optimization in IR systems, covering both evaluation dimensions and corresponding metrics. We categorize existing evaluation approaches into two groups: evaluation for generic preference and personalized preference. For each category, we further discuss the key evaluation objectives, dimensions, and the associated metrics. An overview of the evaluation metrics is summarized in Table 3.

5.1 Evaluation Objectives

In this section, we systematically summarize three key evaluation objectives for generic content generation: (1) quality, (2) trustworthiness, and (3) controllability. For each objective, we outline the main evaluation dimensions along with the commonly used metrics.

5.1.1 ***Quality-Oriented Evaluation***. To evaluate the quality of generated content, the assessment mainly focuses on several aspects: fidelity and personalization.

Fidelity. Fidelity is an important dimension for measuring the quality of images and text generation, since it usually needs to be highly credible to ensure that the generated content can be grounded to specific items (e.g., products, landing pages). Existing studies predominantly adopt distribution-based metrics that compare generated samples with real images in a learned feature space. Specifically, *Inception Score (IS)* and *Fréchet Inception Distance (FID)* are widely used in prior works (e.g., outfit image generation tasks [56, 136, 145, 150], personalized image generation [89, 137], and advertising poster generation [34]). Some studies [45, 66, 136, 137] further adopt *CLIP Score* to measure the semantic alignment between generated images and conditioning text, thereby evaluating whether the generated content faithfully reflects the intended textual description. Inspired by IS, Kumar and Gupta [59] proposes *Search Engine Criteria (SEC)*, which uses the generated images as queries to an image search engine (e.g., Bing Image Search). If the retrieved results contain semantically consistent items (e.g., jeans or pants), the generation is considered of high quality; otherwise, it is regarded as low-quality. More recent studies further employ feature-space discrepancy measures such as *CLIP Maximum Mean Discrepancy (CMMD)* [89], which replaces Inception features with CLIP embeddings and measures the distributional distance using maximum mean discrepancy. Fan et al. [34] further uses *Overlap (Ove)* to measure the overlap area between visual elements in generated posters, and *Maximum IoU (MIoU)* to evaluate the intersection over union between generated layouts and ground-truth layouts, which together reflect local layout rationality and global layout consistency. Xu et al. [136] also conducts human evaluation to assess fidelity by asking annotators to compare the realism and completeness of generated outfits.

For text fidelity evaluation, text similarity metrics are usually used. Mita et al. [81] utilizes *BLEU-4* and *BERTScore* to measure the similarity between the generated text and the reference advertising text. Kamigaito et al. [54] adopts the commonly used *ROUGE-1*, *ROUGE-2*, and *ROUGE-L* metrics to evaluate the matching degree between the generated advertising text and the specified advertising keywords and landing page information. Intersection-based Coverage is also used to measure the matching degree with keywords, titles, and bodies [54]. In text rendering scenarios, Fan et al. [34] uses *Normalized Edit Distance* (NED) to measure the character-level difference between the generated text and the reference text, thereby evaluating the fidelity of text rendering. Qian et al. [93] uses *Visual Script Correlation (VSC)* to evaluate the semantic alignment between selected video clips and the generated script, thereby measuring cross-modal fidelity. Kamigaito et al. [54] and Mita et al. [81] further conduct human evaluation to assess text fidelity, where the former uses *Relevant* judgments to evaluate alignment with the input landing page content and advertising keywords, while the latter evaluates *Faithfulness* by asking whether the generated advertising text is implied by the landing page description.

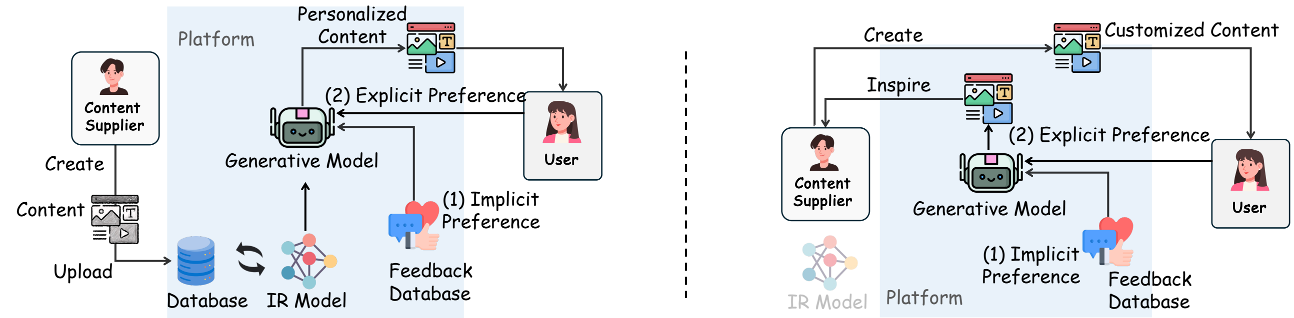
Diversity. Diversity aims to avoid producing overly similar outputs, as this indicates that the generator favors a limited set of modes over others.

For text diversity evaluation, Kamigaito et al. [54] employs the Self-BLEU metric to evaluate the diversity of generated advertising text in search engines. Cui et al. [26] further adopts *Distinct-n (Dist-n)* to quantify the proportion of distinct n-grams in generated texts, where higher values indicate greater lexical diversity.

For image diversity evaluation, previous works [56, 59, 145] uses *Opposite Mean SSIM* to measure the structural dissimilarity between generated images (e.g., fashion image, personalized image) and reference images. Kumar and Gupta [59] proposes the Diversity Criterion (DC) to evaluate the diversity of generated images. For a batch of generated samples, a joint channel-wise color histogram is constructed by aggregating pixel values across all images. The histogram is then normalized to form a probability distribution, and the diversity is quantified by computing the entropy of this distribution.

Aesthetic Quality. In personalized image generation, Lin et al. [66] adopts *Aesthetic Score*, and *PickScore* to evaluate the general quality of generated images. Both metrics assign higher values to images with better aesthetic quality, indicating that the generated images are more likely to be preferred by humans.

Compatibility. In fashion outfit recommendation, recommending visually similar items is often unreasonable, as complementarity is a crucial factor to be considered. [145] utilizes a complementary index, which measures how compatible the generated outfit is by averaging the co-purchase-based compatibility scores between the real-world nearest neighbors of generated items and a validated recommendation item. For complementary fashion generation, Xu et al. [136] uses *Compatibility Score* generated by a compatibility evaluator trained on real-world images to measure whether the generated items are well matched with the target outfit context.



(a) Personalized Generation over Supplier

(b) Personalized Generation for Supplier

Figure 7: Two paradigms of existing personalized content generation for supply-side optimization in IR systems. Personalization is primarily driven by user signals, including explicit feedback (e.g., queries) and implicit feedback (e.g., interaction history), to enhance user experience. (a) Personalization is performed by further refining or adapting content produced by existing suppliers; (b) Personalized content is first generated as inspiration or guidance for suppliers to create customized content.

Text Fluency. Text fluency is often used to evaluate the quality of advertising text generated [54, 55, 81, 123]. Kamigaito et al. [54] employed *Log probability with BERT* to measure the fluency of generated advertising text. Cui et al. [26] uses *Perplexity* (PPL) to assess the fluency and context consistency of dialogue responses in conversational recommendation tasks. Qian et al. [93] uses *Contextual Coherence (Coh)* to evaluate whether the generated script is narratively coherent and contextually smooth. Kamigaito et al. [54] and Mita et al. [81] further conduct human evaluation to assess text fluency, where the former uses *Not Fluent* judgments to evaluate whether generated advertising text is natural and fluent, while the latter directly evaluates *Fluency* by asking annotators whether the text is natural and easy to understand.

Attractiveness. In the text generation tasks in IR systems (e.g., advertising text generation), *attractiveness* is a key metric for evaluating the ability of generated content to capture user attention. Due to its inherently subjective nature, Kamigaito et al. [54], Mita et al. [81] conducts human evaluation to assess *Attractiveness* through pairwise comparison, where annotators judge whether the generated advertising text is more attractive than the human-written reference.

Business Performance. Evaluations also aim to quantify how generated content influences user behavior and commercial outcomes. Existing studies primarily assess attention and engagement through click-based metrics such as click-through rate (CTR), including relative comparisons against baselines (e.g., Win Rate) [23, 34, 142]. To better capture downstream value, more comprehensive evaluations further incorporate conversion-oriented signals such as conversion rate (CVR) and return rate, which reflect whether user interactions translate into actual transactions [66].

In addition, some work considers user-centric proxy indicators of business effectiveness, such as *Product Rating*, *Purchase Intention*, and *Engagement Intention* [79], bridging the gap between immediate interaction signals and longer-term commercial outcomes.

5.1.2 *Trustworthy-Oriented Evaluation*

In addition to the quality of generated content, the credibility of the generation is also an important evaluation aspect for supply-side content generation.

Hallucination. Pretrained generative models contain the potential risk that generated content contains false, fabricated, or misleading information that appears plausible but is not grounded in the provided input or reality. In both text and multi-modal generation, hallucination undermines the reliability and trustworthiness of AI outputs, making its detection and measurement essential for assessing generation quality. *Hallucination Detection Pass Rate* [17] measures the proportion of generated outputs that are judged not to contain hallucinated (false or fabricated) information, with higher values indicating lower false information risk. Qian et al. [93] uses *Logical Correctness (Logic)* to evaluate whether the generated script is factually and logically sound, thereby reflecting its trustworthiness. Beyond hallucination, Ao et al. [5] uses *Clickbait proportion* to measure the proportion of generated news titles that exhibit misleading or exaggerated clickbait characteristics, thereby evaluating the credibility risk of personalized title generation.

5.1.3 *Controllability-Oriented Evaluation*

In practical content generation tasks in IR systems, generated text often needs to meet explicit controllability requirements, such as including specified keywords, adhering to length limits, or following formatting rules.

Keyword Insertion. Mita et al. [81] uses *Keyword Insertion Rate* (KWD) to quantify the extent to which required keywords specified by the task prompt are successfully included in the generated advertising copy.

Length Compliance. It also utilizes *Length Regulation Compliance* (REG) [81] to measure whether the generated copy adheres to prescribed length constraints (e.g., maximum or minimum character/word limits).

Category Correctness. *Inception-based classification accuracy (IS-acc)* [136, 150] is also used to evaluate whether generated images satisfy the conditioning information (e.g., prompts or product attributes). A pretrained Inception classifier predicts the category of generated images, and the classification accuracy with respect to the ground-truth conditions is reported. For example, outfit image generation tasks [136, 150] use IS-acc to measure category correctness.

Table 3: Evaluation metrics for generative supply optimization in IR systems

Dimensions	Metrics	Reference
Image Fidelity	Inception Score (IS), Fréchet Inception Distance (FID)	[34, 56, 89, 136, 137, 145, 150]
	CLIP Score, CLIP Maximum Mean Discrepancy (CMMD)	[56, 89, 136, 145, 150]
	Search Engine Criteria (SEC)	[59]
	Overlap (Ove), Maximum IoU (MIoU)	[34]
Text Fidelity	Bilingual Evaluation Understudy (BLEU), BERTScore, Recall-Oriented Understudy for Gisting Evaluation (ROUGE), Normalized Edit Distance (NED)	[5, 6, 26, 34, 54, 80, 81]
	Visual Script Correlation (VSC)	[93]
	Relevant, Faithfulness	[54, 81]
Diversity	Distinct-n (Dist-n), Self-BLEU (SBLEU)	[26, 54]
	Opposite Mean SSIM, Diversity Criterion (DC)	[56, 59]
Aesthetic Quality	Aesthetic Score, PickScore	[66]
Compatibility	Compatibility Score	[136, 145]
Text Fluency	Log probability with BERT, Perplexity (PPL), Not Fluent, Fluency	[26, 54, 81]
	Contextual Coherence (Coh)	[93]
Attractiveness	Attractiveness	[54, 81]
Business Performance	Click-Through Rate (CTR),Win Rate,Relative CTR Improvement	[20, 23, 34, 66, 69, 80, 113, 141, 142, 158]
	Conversion Rate (CVR), Return Rate	[66, 69, 141]
	Revenue Per Mille (RPM)	[69]
	Product Rating, Purchase Intention, Engagement Intention	[79]
Hallucination	Hallucination Detection Pass Rate	[17, 34]
	Logical Correctness (Logic), Clickbait proportion	[5, 93]
Constraint Satisfaction	Keyword insertion rate (KWD)	[81]
	Length regulation compliance (REG), Average length, Correct length	[54, 81]
	Inception-based classification accuracy (IS-acc)	[136, 150]
	Word Count Discrepancy (WCD), Subtitle Segmentation Accuracy (SSA)	[93]
Personalization	Structural Similarity Index Measure (SSIM), Learned Perceptual Image Patch Similarity (LPIPS)	[71, 124, 137, 140]
	CLIP Image Score (CIS), DINO Image Score (DIS)	[124, 137]
	CLIP Personalization Score (CPS), CLIP Personalization Image Score (CPIS)	[71]
	Verifier Score	[89]
	PerFusionRM Score	[66]
	Average ranks	[124]

Format Compliance. *Word Count Discrepancy (WCD)* [93] measures the absolute difference between the word count of the generated script and the target word count expected for the video segment, reflecting whether the script length matches the segment duration. *Subtitle Segmentation Accuracy (SSA)* [93] measures whether subtitle segmentation satisfies predefined formatting and linguistic constraints, reflecting the readability and presentation quality of generated subtitles.

5.1.4 ***Personalization-Oriented Evaluation***.

Personalized evaluation is crucial for assessing whether the generated content aligns with the preferences of individual users. We summarize the existing approaches into two dimensions: objective and subjective evaluation.

Objective. In personalized content generation tasks [71, 137], *Structural Similarity Index Measure (SSIM)* [71, 137] evaluates the similarity between generated images and reference images by comparing their structural information. *Learned Perceptual Image Patch Similarity (LPIPS)* [124, 137] computes perceptual differences based on deep feature representations and aligns better with human visual perception. *CLIP Image Score (CIS)* and *DINO Image Score (DIS)* [124, 137] are adopted to measure the visual similarity between generated images and user history images or reference images based on CLIP and DINO features, respectively, which are useful for evaluating personalized alignment such as semantic consistency and style preservation. Meanwhile, *CLIP Personalization Score (CPS)* and *CLIP Personalization Image Score (CPIS)* [71] further assess personalization by measuring the similarity of generated images to user preference representations in the text and image modalities, respectively.

In addition to these offline metrics, online metrics can also be used for personalized content evaluation. For example, Lin et al. [66] reports online A/B testing results using business-oriented metrics such as *CTR*, *CVR*, and *Return Rate* to evaluate whether personalized generated fashion images better match users' preferences in real-world e-commerce scenarios. In conversational and generative recommendation settings, Cui et al. [26] further validates personalization effectiveness through online click-related metrics, showing that personalized generation and recommendation can improve user engagement in deployed systems. Similarly, Xi et al. [130] conducts online A/B testing on real-world news and music recommendation platforms, and uses metrics such as *recall rate, play* *count*, and *watch time* to evaluate whether knowledge-enhanced recommendations better satisfy users' personalized interests.

Subjective. Some studies even adopt task-specific models (e.g., trained reward models) or human evaluation to evaluate personalized generated content. For example, Patron [89] uses a user-image preference prediction model to compute *Verifier Score*, which is used to measure the personalization strength of generated results. In personalized image generation, Lin et al. [66] adopts *PerFusionRM Score* to evaluate the quality of generated images from the perspective of user preference. Wang et al. [124] uses *Average ranks* to evaluate how well the generated results align with user preferences by asking human evaluators or LLM judges to rank outputs from different models. Xu et al. [136] conduct human evaluation to assess personalization by asking annotators to compare or rank generated results according to how well they align with users' historical preferences.

5.2 Evaluation Methodology

In this section, we focus on evaluation methodologies for generative content in IR systems. We categorize existing approaches into three main paradigms: offline automatic evaluation, human and LLM-as-a-judge evaluation, and online evaluation.

5.2.1 ***Offline Evaluation.*** Offline evaluation provides a set of automatic metrics to assess the quality of generated content without real user interaction, and is widely used due to its efficiency and reproducibility. Existing methods can be broadly categorized based on their evaluation objectives and reference signals.

For image generation, offline metrics mainly focus on distribution-level fidelity and instance-level similarity. Distribution-based metrics such as Inception Score (IS) [9], Fréchet Inception Distance (FID) [34, 56, 89, 136, 137], and CMMD [89] evaluate how closely generated images match real data distributions or conditional semantics. Instance-level similarity metrics, such as SSIM [71, 137] and LPIPS [124, 137], measure structural and perceptual similarity to reference images. In addition, feature-based semantic metrics, including CLIP Score [34, 66] and DINO-based scores [66, 121, 136], assess cross-modal alignment, while variants such as CPIS [71] extend them to personalization settings. Structural metrics such as mIoU and overlap ratio [34] further evaluate layout consistency.

For text generation, offline evaluation relies on reference-based similarity metrics such as BLEU, ROUGE, and normalized edit distance (NED), which measure n-gram overlap with gold references [5, 81]. Beyond similarity, hallucination metrics such as hallucination detection pass rate [17] assess factual consistency and reliability. Diversity is captured by Dist-n [26] and Self-BLEU [54], while constraint satisfaction is evaluated via keyword insertion rate (KWD) and length regulation compliance (REG) [81].

Overall, offline evaluation enables scalable and standardized assessment of generative models, but it primarily captures proxy signals and may not fully reflect user perception or real-world business impact.

5.2.2 ***Human/LLM as Judge.*** A more reasonable way to assess whether generated content aligns with human preferences is to rely on more subjective evaluation methods.

Human evaluation. Human evaluation is widely adopted to better capture users' perceived quality of generated content. Existing studies typically assess whether generated content aligns with user needs or contextual conditions through dimensions such as *Compatibility*, *Fidelity*, and *Personalization* [106, 136].

In advertising-related generation, human judgments further evaluate content effectiveness and user response, including *Attractiveness*, *Relevance* [54], *Faithfulness*, and *Fluency* [81], as well as downstream business indicators such as *Product Rating*, *Purchase Intention*, and *Engagement Intention* [79].

Beyond quality assessment, human evaluation is also employed to measure practical risks and usability, such as *clickbait proportion* in personalized news generation [5] and *accuracy* in e-commerce product information repair [78].

LLM as Judge. Recently, LLM-as-judge has emerged as an effective paradigm for evaluating the quality of AIGC [43] [43]. In IR ecosystems, Lin et al. [69] uses an LLM to score generated content from the dimensions of helpfulness, relevance, and accuracy. In advertising video generation, Qian et al. [93] employs an LLM to evaluate content through *Visual Script Correlation* (VSC), *Contextual Coherence* (Coh), and *Logical Correctness* (Logic), focusing on whether the generated video content is well aligned with the script and logically coherent. In personalized image generation, Wang et al. [124] uses the LLM's *average ranks* over candidate outputs to compare different methods from a relative preference perspective.

5.2.3 ***Online Evaluation.*** Online evaluation focuses on business-oriented metrics to assess whether generated content improves real-world performance under live user traffic. These metrics are typically measured via online A/B testing [58], where generated content is deployed to different user groups and its impact is evaluated on key indicators such as click-through rate (CTR), conversion rate (CVR), and revenue.

Existing studies predominantly adopt CTR-based evaluation to measure the effectiveness of generated content in attracting user attention. For example, prior work evaluates the CTR performance of generated creatives in real e-commerce advertising systems [34, 142], while others further compare generated content with human-written baselines using metrics such as Win Rate [23].

Beyond click-based metrics, more comprehensive evaluation protocols incorporate downstream signals such as CVR and return rate to better capture true business value and mitigate the risk of optimizing for clicks without improving conversions [66].

6 Challenges of Content Generation

We provide a detailed discussion of the challenges of content generation in two-sided IR systems, highlighting key issues and their impact on both the content ecosystem and personalized experience in IR systems.

6.1 Content Ecosystem Challenges

Generic content generation primarily aims to optimize content supply for providers, improving exposure, user engagement metrics (e.g., CTR), and revenue. However, these supplier-oriented objectives may not align with user preferences. Moreover, biases and hallucinations in generative models can be amplified through IR system feedback loops, potentially distorting retrieval outcomes and degrading user experience.

6.1.1 ***IR Engine Poisoning Risks.*** Recent IR systems predominantly rely on retrieving content from a shared content corpus, including both traditional ranking-based paradigms [12, 72, 162] (which directly present ranked items to users) and RAG pipelines [63]. The growing reliance of IR systems on generative content within item corpora and generative IR architectures introduces new security vulnerabilities that can be exploited by adversarially crafted AI-generated content (AIGC).

Recent studies have shown that traditional and generative IR pipelines are vulnerable to data and knowledge poisoning attacks, where adversaries inject malicious content into external knowledge bases to manipulate retrieval and generation behavior [61, 117, 163]. These attacks typically operate by inserting adversarial or AI-generated documents into the corpus, causing neural retrievers to misjudge relevance and elevate attacker-controlled content in ranked results or generated responses [109]. Notably, even a small number of injected documents can significantly bias system outputs, and such effects can be further amplified through query-aware adversarial content or joint optimization over retrieval and generation stages [61, 117, 163]. Beyond content poisoning, related threats include membership inference and semantic memory poisoning, which can lead to sensitive information leakage or long-term corruption of external knowledge bases [24]. In addition, adversaries may embed prompt-style instructions into documents to hijack citation behavior and bias model outputs toward attacker-controlled sources [84].

6.1.2 ***Hallucination and Misinformation.*** Generative models enable large-scale, automated production of misinformation by generating high-quality, semantically coherent multi-modal content, including text, images, audio, and videos (e.g., deepfakes), that is often difficult to distinguish from human-authored information [25, 86]. This capability substantially lowers the cost of misinformation creation while facilitating high-frequency, audience-targeted dissemination and increasing the realism and deceptiveness of fabricated content [86]. When propagated through IR ecosystems, such content can spread rapidly and at scale, exceeding the capacity of traditional human-centered moderation and amplifying the risks associated with AI-generated hallucinations [31].

Motivated by these concerns, a growing body of work investigates the real-world harms of AI-generated hallucinations, particularly in mass media and public information environments [77, 87, 104]. These studies show that misleading or fabricated AIGC (e.g., fake news) can inflict significant harm on individuals, institutions, and the public sphere [77, 87]. As AIGC becomes increasingly embedded in information ecosystems, the scale and severity of such risks are likely to intensify.

From a platform perspective, widespread exposure to misleading content can erode user trust, undermining long-term user retention and platform sustainability [46]. From a user perspective, repeated exposure to misinformation can reshape behavioral patterns, increasing the likelihood that users attend to, click, like, or share low-credibility content while reducing engagement in fact-checking behaviors [8, 31]. Even a single exposure to false information can alter real-world intentions and decisions, for example by reducing willingness to adopt certain health-related behaviors [10, 42]. At the societal level, the spread of AIGC-driven misinformation can have particularly severe consequences in crisis contexts, including declines in public safety and collective trust [76, 114].

6.1.3 ***Social Bias and Unfairness.*** Prior studies have shown that modern pretrained generative models often encode substantial social biases and unfairness [35]. As AIGC becomes increasingly integrated into IR ecosystems, such biases may propagate from model outputs to downstream content generation and recommendation processes, leading to systematically unequal user experiences and raising concerns about group- and individual-level fairness [125].

Empirical studies further demonstrate that large language models (LLMs) may produce systematically different outputs for users across demographic attributes such as age, gender, and nationality [48]. These disparities are largely rooted in the pretraining process, where large-scale web corpora inevitably contain historical, societal, and cultural biases. Such biases can be encoded during training and further amplified through generation and iterative deployment in IR systems [35, 75, 156].

In addition, within IR systems, these biases may not remain isolated at the content generation stage. Instead, they can be reinforced through feedback loops between user interactions and ranking models, potentially resulting in skewed exposure distributions, unequal information access, and long-term fairness degradation in the content ecosystem.

6.1.4 ***Content Ecosystem Homogenization.*** A key challenge introduced by large-scale generative content production in IR systems is the risk of content ecosystem homogenization. While generative models enable scalable and personalized content creation [64, 122], they may also inadvertently reduce content diversity by favoring high-probability, pattern-consistent, or engagement-optimized outputs [53, 57, 108]. As a result, generated content across different users, queries, or contexts may become increasingly similar in structure, style, and semantics.

Such homogenization is further amplified in two-sided IR systems due to feedback loops between generation and recommendation. Content that performs well under existing ranking and engagement signals is more likely to be generated and amplified in the future, leading to a "rich-get-richer" dynamic at the content level. Over time, this may reduce exploration of novel or long-tail content, thereby weakening ecosystem diversity and limiting user exposure to diverse information.

From a system perspective, this trend raises concerns not only for user experience but also for the long-term sustainability of the content ecosystem, as reduced diversity may lead to declining user engagement, creator incentives, and overall informational value.

6.1.5 ***Source Bias.*** With the large-scale integration of AIGC into the IR ecosystem, retrieval systems increasingly operate over corpora where human-generated content (HGC) and AI-generated content coexist. Recent studies have identified a phenomenon termed source bias [27, 28], whereby neural retrieval models disproportionately favor LLM-generated documents over human-written ones, even when their semantic relevance is comparable.

To quantify this phenomenon, Dai et al. [28] proposes metrics such as Relative Δ to measure both the direction and magnitude of the bias. Furthermore, Wang et al. [115] attribute this bias to the fact that PLM-based retrievers implicitly exploit textual perplexity as an auxiliary signal, while LLM-generated content typically exhibits lower perplexity. Importantly, this bias is not limited to textual retrieval. Recent work shows that source bias also emerges in cross-modal retrieval settings: incorporating AI-generated images into training data can shift ranking preferences toward generated images [135], and video retrieval models may similarly prefer AI-generated videos over real content [37].

This phenomenon poses a significant challenge because it can distort retrieval results, amplify exposure disparities, and compromise fairness and content diversity [27, 28, 135]. When retrieval models systematically favor AI-generated content, relevant human-authored documents may be ranked lower, leading to reduced visibility and skewed information access for users. Moreover, source bias may propagate through feedback loops in the IR ecosystem. Initial ranking preferences influence content exposure and user interactions, which subsequently shape future training data and model updates. Such dynamics can progressively reinforce the preference toward AI-generated content and further marginalize human-authored content [21, 160]. Over time, this self-reinforcing process may reduce content diversity, distort the information landscape, and weaken the sustainability of mixed-source content ecosystems. These dynamics make source bias a persistent and compounding challenge in modern retrieval systems where human-generated and AIGC coexist.

6.2 Personalization Challenges

Personalized generation faces two key challenges. First, evaluation remains difficult, as it requires assessing user-specific utility, yet lacks reliable offline evaluation methods. Second, there exists a fundamental trade-off between fidelity and personalization, where increasing personalization may deviate from the original content, potentially affecting consistency and reliability.

6.2.1 ***Evaluation Difficulty.*** Despite the growing popularity of personalization techniques, there remains a clear lack of standardized benchmark datasets and reliable evaluation metrics that can accurately assess the performance of different approaches. In the evaluation of personalization, human evaluation provides reliable preference signals. However, it introduces significant challenges in terms of scalability, cost, and reproducibility. Currently, content personalization is often evaluated using metrics (e.g., CLIP-based image similarity [122]). However, when models overfit to reference data, such metrics may overestimate performance and fail to reflect true generalization ability. To address these limitations, recent studies have increasingly adopted LLMs as judges to evaluate creative outputs [43, 93]. Nevertheless, concerns remain regarding the reliability, bias, and fairness of such approaches.

Therefore, future research should focus on developing comprehensive and widely accepted benchmarking frameworks to evaluate personalized content generation models across multiple dimensions, including but not limited to alignment with specific user preferences and aesthetic quality.

6.2.2 ***Trade-off of Fidelity and Personalization.*** The ultimate goal of personalized content generation in supply-side optimization is to build systems that can not only faithfully preserve the identity of a target subject but also effectively respond to diverse personalized textual prompts (e.g., user profile). However, achieving both objectives simultaneously remains inherently challenging due to a fundamental trade-off.

On the one hand, high subject fidelity requires capturing and reconstructing fine-grained and distinctive characteristics of the target subject, which often encourages models to minimize reconstruction loss and faithfully replicate subtle visual details. On the other hand, personalized text alignment demands flexible adaptation of the subject according to varying textual descriptions, which may involve changes in pose, expression, background, or stylistic attributes, rather than strict adherence to the reference appearance.

Therefore, while improving fine-grained subject preservation, enabling flexible adaptation across diverse contexts remains a key challenge. More effective model architectures, novel training strategies, and dynamic data modeling techniques are still needed to better balance subject fidelity and personalized text faithfulness in personalized content generation systems.

7 Future Directions

In this section, we outline several promising future research directions of content generation for supply-side optimization in IR systems, aiming to inspire researchers to better understand and advance generative supply-side optimization in IR systems.

7.1 Generative Mechanism Design for Supply-Side Optimization

Supply-side optimization not only involves the direct improvement of content itself, but can also influence the strategic behavior of content providers through mechanism design, thereby indirectly shaping content quality and distribution. Prior work has shown that two-sided traffic allocation mechanisms can incentivize providers to adjust their creation strategies to some extent. However, such approaches primarily rely on traffic-based signals (e.g., exposure or clicks) as incentives, which limits their effectiveness, since traffic constitutes only one of many feedback signals perceived by content providers.

To further enhance the ability to influence creator behavior, some studies [92] in economics have proposed leveraging richer information design mechanisms (e.g., strategic prompting or information disclosure) to guide the creator's creation behavior. Inspired by theories of information disclosure, these approaches aim to more directly affect decision-making by carefully designing the signals or feedback presented to content providers. For instance, prior work has explored reinforcement learning-based strategies for optimal information revelation, and evaluated their impact on creator behavior through simulation.

These studies offer an important insight: generative models can be used not only to directly optimize content, but also to generate customized signals or feedback that influence the behavior of content providers. In doing so, they enable a deeper level of intervention in the content ecosystem, potentially guiding it toward more desirable properties such as diversity, quality, and sustainability.

7.2 Personalized Generation as Supplier

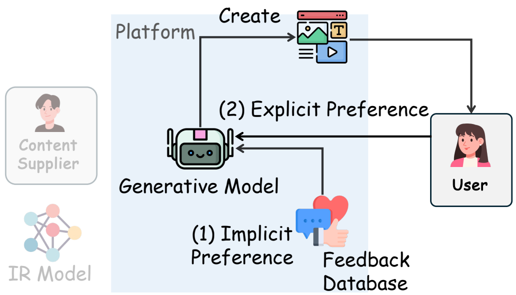


Figure 8: The generative model directly acts as the content supplier, producing personalized content for users without relying on content suppliers or IR models.

In generative supply-side optimization, existing personalized content generation techniques are typically not independent of content suppliers. As illustrated in Figure 7, prior methods either generate content based on item produced by suppliers (Figure 7 (a)), or provide personalized signals to guide suppliers in producing customized content (Figure 7 (b)). However, recent studies [121] have proposed a new paradigm that directly leverages generative models for personalized multimodal content creation without relying on any explicit content supplier, as illustrated in Figure 8.

We refer to this paradigm as *personalized generation as supplier*, where the generative model itself plays the role of the content supplier. This approach fundamentally replaces the traditional pipeline in which suppliers first create content, and IR systems subsequently perform distribution. Instead, it directly enables end-to-end personalized content generation through generative models.

7.3 Agentic Content Generation

Existing content generation approaches primarily rely on direct generation [52, 54, 55, 60, 123] , self-iterative [158], or multi-stage pipelines [32, 39, 116, 126, 142] powered by generative models (e.g., LLMs) to conduct content generation. While effective, such approaches often lack generalizability across scenarios, requiring task-specific prompts, reward designs, or manual adaptations when transferred to new settings.

Beyond acting as mere generation tools in the creation process, generative models can serve as intelligent agents that actively acquire external knowledge and orchestrate diverse tools. Recent advances in LLM-powered agents [19, 74, 131, 148] enable the integration of modular skills, allowing models to autonomously plan, retrieve information, and invoke specialized tools [94, 118] to accomplish complex tasks.

Content creation, in practice, is inherently multi-step and tool-dependent. For example, an agent may first leverage external knowledge retrieval (e.g., deep research agent [51, 152, 154]) to gather relevant information, then utilize coding tools [30, 153], GUI interface [68, 85], or other domain-specific tools for structured processing, and finally generate content conditioned on both user context and retrieved knowledge. Such agentic workflows offer a promising direction toward more flexible, generalizable, and scalable content generation in IR systems.

7.4 Joint Optimization of Content Generation and Matching

A key future direction is to move beyond the current decoupled design of supply-side optimization and demand-side matching, and instead investigate their tight interaction in two-sided IR systems. Existing approaches typically optimize content generation (supply side) and retrieval or ranking (demand side) separately, which may lead to suboptimal system-level outcomes due to misaligned objectives and feedback gaps between the two sides.

Future work may explore closed-loop frameworks in which content generation and matching are jointly modeled as an interactive process. In such settings, generated content is influenced by downstream matching signals, while matching policies adapt to the evolving content supply. This bidirectional interaction has the potential to enable more coherent system-level optimization, improving not only user experience but also creator incentives and the long-term sustainability of the content ecosystem.

8 Conclusion

This survey systematically examines content generation for supply-side optimization in two-sided IR systems, highlighting its importance beyond traditional IR paradigms centered on supply–demand matching. While conventional approaches primarily focus on improving the allocation of existing content, content generation shifts the focus toward the creation and refinement of content itself, which plays a fundamental role in shaping user experience as well as the overall health of the content ecosystem.

To better structure this emerging field, we present a unified taxonomy covering generative supply-side optimization methods, evaluation paradigms, and key challenges. By organizing the literature along these dimensions, this survey provides a coherent and comprehensive perspective on a rapidly evolving research area, and aims to inspire future work toward more effective, robust, and trustworthy generative supply-side optimization in IR systems.

Appendix

A Positioning with Existing Surveys

Table 4 presents a comparison between our survey and representative prior surveys on generative models for information retrieval (IR) and related applications. Existing works primarily focus on specific application domains or modeling paradigms, such as video generation [159], personalized generation with LLMs [139], generative search [63], and generative or multimodal recommendation [72, 73, 120]. In addition, some studies investigate particular model families, including LLM-based methods and diffusion models, within specific tasks [65, 161].

Despite these advances, existing surveys typically remain limited along two key dimensions. First, they largely focus on specific tasks (e.g., search or recommendation) and do not provide a unified view across multiple IR scenarios, such as search, recommendation, and advertising. Second, most prior work emphasizes model architectures or application pipelines, while overlooking the role of content generation as a supply-side optimization mechanism in two-sided IR systems.

In contrast, our survey provides a comprehensive and unified perspective on content generation in IR systems. As shown in Table 4, we systematically cover multiple IR scenarios, including search, recommendation, and advertising, while incorporating both LLM-based and diffusion-based generation paradigms. More importantly, we explicitly model IR systems from a supply-side perspective, highlighting how content generation reshapes the content ecosystem and interacts with demand-side matching.

Table 4: Comparison with existing surveys on generative models for information retrieval and related applications

Survey	Supply side	Search	Recommendation	Advertising	LLM	Diffusion Model	Research Problem
Zhou et al. [159]	✗	✗	✗	✗	✓	✗	Video generation
Xu et al. [139]	✗	✗	✗	✗	✓	✗	Personalized generation
Li et al. [63]	✗	✓	✗	✗	✓	✗	Generative information retrieval
Lin et al. [65]	✓	✗	✓	✗	✗	✓	Diffusion model for recommendation
Liu et al. [73]	✗	✗	✓	✗	✓	✓	Multi-modal recommendation
Liu et al. [72]	✗	✗	✓	✗	✓	✗	LLM-based recommendation
Zhu et al. [161]	✗	✗	✓	✗	✓	✓	Multi-modal recommendation
Wang et al. [120]	✗	✗	✓	✗	✓	✓	Generative recommendation
Ours	✓	✓	✓	✓	✓	✓	Content generation in IR systems

B Trends in Content Generation Research for IR Systems

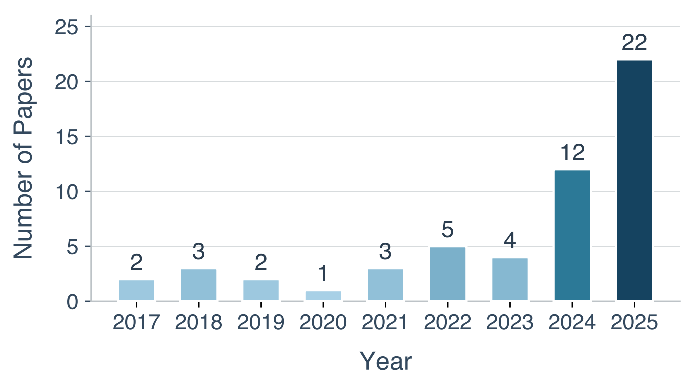


Figure 9: Yearly publications on content generation in IR systems (2017-present).

As shown in Figure 9, the trend of content generation research in information retrieval systems has experienced rapid growth in recent years, driven by the continuous advancement of generative models. Based on our systematic review of the literature since 2017, we observe a clear and sustained acceleration in research activity.

Notably, the release of SORA in 2023 marked a pivotal turning point, significantly lowering the barrier to deploying generative models in real-world applications. This was followed by the emergence of powerful multimodal generative models, such as *NaNo Banana*, *GPT-Image-2*, which further expanded the capability of content generation.

As a result, content generation has evolved from a complementary technique to a core component of modern IR systems, and research on generative methods has exhibited an exponential growth trend.

C Details of Generative Supply-Side Optimization

C.1 Generic Content Generation

Table 5: Details of recent research on generic content generation.

Paper	Information Required	Brief Description	Year
		Advertising Text Generation	
[80]	Source advertising text, image	CTR-oriented advertising creatives refinement with multi-advertiser and paired A/B testing data	2020
[55]	Landing page info	Diverse search engine text advertisement generation with a Seq2Seq model [90] and self-critical sequence training	2021
[60]	Product info, prompt	Advertising text generation via three-phase transfer learning and multi-task prompt learning	2022
[127]	User review, Aspect term	CTR-oriented advertising text generation via controlled pre-training and contrastive fine-tuning	2022
[158]	History copy	Text advertisement refinement via LLM with genetic evolution of prompts	2024
[70]	Aspect term, Review, Queries	CTR-oriented review-based copywriting generation via multi-reward reinforcement learning	2024
[54]	Landing page info, Ad keywords	Search engine text advertisement generation conditioned on landing page semantic reuse	2025
[23]	Product info	CTR-oriented advertisement copy generation via LLM-based online preference learning	2025
[113]	Original copy	CTR-oriented marketing copy generation via LLM-based iterative refinement	2025
		Advertising Image Generation	
[110]	Face image, Category	Persuasion-aware, category-specific advertising facial image generation	2018
[67]	Original poster, Poster text	Product poster generation with content-aware layout, automatic slogan generation, and multi-attribute style prediction	2023
[38]	Product background, Poster text	Product poster text rendering via background-aware styling and text-semantic-guided emphasis	2023
[62]	Product background, Poster text	Product poster generation via visual-textual relation modeling and geometry-aware diffusion	2023
[14]	Product cutout, product name	Product background generation with prompt-free diffusion and MLLM-guided iterative refinement	2024
[119]	Prompt	Chinese poster generation with background modeling, layout control, and graphical text synthesis	2024
[112]	Product name	Advertisement background generation via preference-aware prompt construction for diffusion synthesis	2024
[32]	Prompt, Product image	Advertising background generation via diffusion model with feedback-guided refinement	2024
[142]	Product info	Advertising creative image generation via diffusion models with CTR-oriented optimization	2024
[126]	Prompt, Logo image	Advertising banner generation with multi-agent iterative refinement	2025
[116]	Prompt	Code-level advertisement banner generation via multi-agent cooperation	2025
[4]	Prompt, Style image	Culture-aware and bias-controlled advertising image generation	2025
[34]	Product information	MLLM and diffusion-based advertising poster multi-stage generation	2025
[39]	Product foreground, Poster text, Prompt	Product poster generation via character-discriminative visual features and subject fidelity feedback learning	2025
[16]	Product foreground, Poster text	Product poster generation via layout planning and layout-controlled inpainting	2025
[20]	Product info	CTR-oriented advertising background generation via prompt-guided diffusion	2025
		Advertising Video Generation	
[93]	Product info, Video clips	Video advertisement generation via LLM-based segment selection and script generation	2025
[2]	Item title	Structured item description generation via LLM	2023
[78]	Item info	Product attribute repair and completion via constraint-guided LLM tool-calling generation	2025
		Generative Engine Optimization	
[3]	Web content, User query	Foundational generative engine optimization framework that boosts citation chance in LLM-based search	2024
[129]	Web content, User query	Extracts GE preferences via LLMs and builds prompt/RL-based models for web content optimization	2025
[151]	Web content, User query	Self-evolving GEO agent with archive-driven strategy selection and critic-guided adaptive content rewriting	2026

C.2 Personalized Content Generation

Table 6: Details of recent research on personalized supply-side optimization.

Paper	Information Required	Brief Description	Year
		Personalized Generic Image Generation	
[105]	User behaviors, Item image	Personalized item image generation via diffusion models with preference embeddings extracted from LLMs	2024
[124]	User interactions, Item image	Personalized item image generation via diffusion models with LLM-rewritten user interests	2024
[89]	User ID, Desired rating	Learn user-conditioned image embeddings via diffusion prior and decode personalized images with generative model	2025
[137]	User interactions, Reference image	Personalized image generation via self-supervised reconstruction alignment with user preference tokens	2025
[71]	User interactions, Reference image	Personalized image generation via recommendation signals and relevant history retrieval	2025
[140]	User interactions, Reference image	Personalized image generation with disentangled semantics and style fused with user preferences	2025
[122]	User history, Target item image	Personalized item image generation using preference-guided diffusion in conversational recommendation	2025
		Personalized Outfit Image Generation	
[56]	User embedding, Item category	Personalized outfit image generation via GAN conditioning with user-aware visual embeddings	2017
[59]	Item image	Top-to-bottom compatibility outfit image generation via conditional GAN training	2017
[145]	Item image	Compatible outfit image generation via GAN-conditioned signals with user preference embeddings	2018
[106]	Item image	Compatible outfit image generation via compatibility-aware prototype-conditioned GANs	2018
[26]	User interaction title	Personalized fashion item title generation with a multi-modal recommendation foundation model	2022
[136]	User interaction image, Incomplete outfit image	Personalized outfit image generation via diffusion models with user-history conditioning	2024
[66]	User query	Personalized text-to-image generation via reward-guided diffusion and user preference conditioning	2025
[150]	Incomplete outfit image	Outfit image generation via diffusion models with preference learning and multi-expert DPO feedback	2025
		Personalized Text Generation	
[5]	User interactions, Candidate news	Personalize headline generation via user-interest-aware pointer-generator	2024
[17]	Advertisement content, User info	Hierarchical LLM-driven personalized advertisement title generation	2025
		Personalized Video Generation	
[138]	User profile, Target product, Context	Personalized, context-aware video advertisement generation with multi-agent	2026
		Search Advertisement Generation	
[52]	Landing page info, User query	Search engine text advertisement generation with parallel encoder-decoder and self-critical sequence training [99]	2019
[123]	Landing page info, User query	Search engine text advertisement generation with pretrained model-based reinforce learning	2021
[41]	Landing page info, User query	Search engine text advertisement generation with query-conditioned customization and factuality-aware modeling	2022
		On-Demand Product Generation	
[66]	User query	Personalized text-to-image generation via reward-guided diffusion and user preference conditioning	2025

D Evaluation Metric Details

Table 7: Evaluation metric details of visual content generation in IR systems.

Metric	Description
Inception Score (IS)	*Inception Score (IS)* employs a pre-trained Inception-v3 [9] classification model to evaluate whether generated images are (1) clearly recognizable (i.e., the predicted class distribution is concentrated) and (2) diverse across different categories. It reflects both the sharpness and diversity of generated images.
Search Engine Criteria (SEC)	*Search Engine Criteria (SEC)* submits the generated images as queries to an image search engine (e.g., Bing Image Search). If the retrieved results contain semantically consistent content (e.g., jeans or pants), the generated result is considered high-quality; otherwise, it is regarded as low-quality.
Fréchet Inception Distance (FID)	*Fréchet Inception Distance (FID)* measures how close generated images are to real images [34, 56, 89, 136, 137]. Specifically, it assumes that the feature representations of real and generated images follow Gaussian distributions and computes the 2-Wasserstein distance between these two distributions.
CLIP Maximum Mean Discrepancy (CMMD)	*CLIP Maximum Mean Discrepancy (CMMD)* [89] compares feature-space distributions between generated and real images under specific attribute conditions, thereby measuring semantic consistency at the conditional distribution level.
Diversity Criterion (DC)	For a batch of generated images, a joint channel-wise color histogram is constructed by aggregating pixel values across all images. The histogram is then normalized to form a probability distribution. The diversity is quantified by computing the entropy of this distribution. A higher entropy indicates greater diversity among the generated images.
BRISQUE	*Blind/Referenceless Image Spatial Quality Evaluator (BRISQUE)* is a no-reference image quality assessment metric that evaluates perceptual naturalness based on spatial scene statistics, without requiring a reference image. Lower values usually indicate better perceptual quality.
Structural Similarity Index Measure (SSIM)	*Structural Similarity Index Measure (SSIM)* [71, 137] evaluates the similarity between generated images and reference images by comparing their structural information. It considers luminance, contrast, and structural patterns, thereby emphasizing the preservation of global structures rather than pixel-wise differences.
Learned Perceptual Image Patch Similarity (LPIPS)	*Learned Perceptual Image Patch Similarity (LPIPS)* [124, 137] computes perceptual differences based on deep feature representations and aligns better with human visual perception. These metrics are widely used in personalized generation [71, 137], image editing, and style transfer tasks.
CLIP Image Score (CIS)	*CLIP Image Score (CIS)* measures the similarity between generated images and reference images in the CLIP feature space, and is mainly used to assess high-level semantic consistency.
DINO Image Score (DIS)	*DINO Image Score (DIS)* measures the similarity between generated images and reference images in the DINO feature space, and is often used to capture visual consistency from self-supervised vision representations.
CLIP Personalization Image Score (CPIS)	Based on the CLIP image score, Ling et al. [71] proposed *CLIP Personalization Image Score (CPIS)* [71] to evaluate the similarity between generated images and user-preferred images for personalization assessment.
Overlap (Ove)	*Overlap (Ove)* measures the area of overlap between any two elements in a generated poster, and is used to evaluate local layout quality by reflecting whether the layout suffers from element occlusion or improper placement. Lower values indicate clearer element arrangement and better local layout rationality [34].
Maximum IoU (MIoU)	*Maximum IoU (MIoU)* evaluates the intersection over union between the generated layout and the ground-truth layout, and is used to assess global layout consistency. Higher values indicate better agreement between the generated layout and the reference layout [34].
Opposite Mean SSIM	*Opposite Mean SSIM* [59] measures the structural dissimilarity between generated images and reference images. Higher values indicate greater structural divergence, while lower values indicate closer structural similarity. It is often used in tasks such as style transfer or personalized image generation, where diversity or deviation from the reference is desirable.
CLIP Score	*CLIP Score* [34, 66] measures text–image alignment by computing the similarity between generated images and their corresponding textual descriptions in the CLIP embedding space.
CLIP Personalization Score (CPS)	Based on CLIP Score, Ling et al. [71] proposed *CLIP Personalization Score (CPS)* to evaluate whether the generated image is aligned with the user description text for personalization assessment.
Inception-based Classification Accuracy (IS-acc)	*Inception-based classification accuracy (IS-acc)* [136, 150] evaluates whether generated images satisfy the conditioning information (e.g., prompts or product attributes). A pretrained Inception classifier predicts the category of generated images, and the classification accuracy with respect to the ground-truth conditions is reported. For example, outfit image generation tasks [136, 150] use IS-acc to measure category correctness.

Table 8: Evaluation metrics of image content generation in IR systems.

Metric	Description
Bilingual Evaluation Understudy (BLEU)	*Bilingual Evaluation Understudy (BLEU)* measures the similarity between generated text and reference text based on n-gram overlap, and is widely used in tasks with explicit references, such as item title generation and advertising copy generation [81].
Recall-Oriented Understudy for Gisting Evaluation (ROUGE)	*Recall-Oriented Understudy for Gisting Evaluation (ROUGE)* measures the overlap between generated text and reference text from a recall-oriented perspective, and is commonly used to evaluate how well generated content covers the key information in the target text [5, 54].
Normalized Edit Distance (NED)	*Normalized Edit Distance (NED)* measures the character-level difference between generated text and reference text based on edit distance, and is particularly useful in text rendering scenarios such as poster generation [34].
Self-BLEU (SBLEU)	*Self-BLEU* (SBLEU) [54] evaluates the similarity between multiple generated outputs by treating each generated text as a candidate and the remaining outputs as references. Lower SBLEU scores indicate greater diversity among the outputs, while higher scores suggest that the model tends to produce repetitive or template-like text. Together with Dist-n, SBLEU provides a complementary measure of inter-output diversity.
Distinct-n (Dist-n)	*Distinct-n* (Dist-n) [26] measures the proportion of unique n-grams in the generated texts, providing an estimate of the internal diversity and repetition within each output. A higher Dist-n value indicates that the text contains more varied expressions and fewer repeated phrases, which is desirable in tasks such as advertising copy or recommendation title generation.
Keyword Insertion Rate (KWD)	*Keyword Insertion Rate* (KWD) quantifies the extent to which required keywords or key phrases specified by the task prompt or instruction are successfully included in the generated text. It is often used in tasks such as advertising copy generation where certain lexical elements must appear to satisfy semantic or marketing requirements [81].
Length Regulation Compliance (REG)	*Length Regulation Compliance* (REG) [81] measures whether the generated text adheres to prescribed length constraints (e.g., maximum or minimum character/word limits). In many applications such as ad title or product description generation, output texts must fall within strict length ranges to meet display or usability requirements, and REG captures the proportion of outputs that satisfy these constraints.
Hallucination Detection Pass Rate	*Hallucination Detection Pass Rate* [17] measures the proportion of generated outputs that are judged not to contain hallucinated (false or fabricated) information, with higher values indicating lower false information risk.
Correct Length	*Correct Length* measures the percentage of generated advertising texts that do not exceed the prescribed length limit.
Average Length	*Average Length* measures the average length of generated advertising texts. In general, values closer to the length limit are preferred, as long as the generated text does not exceed the limit.
Log probability with BERT (Log Prob.)	*Log probability with BERT (Log Prob.)* uses the prediction probabilities of a masked language model to estimate the fluency of generated text. In practice, this metric can be computed with models such as bert-base-japanese-v2 in HuggingFace Transformers, following the masked language modeling setting.

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